Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 108–118

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

IndEmoVis: A Multimodal Repository for In-Depth Emotion Analysis in Conversational Contexts

Ruhina Karani [a,*] , Dr. Sharmishta Desai [b]

[a]*Research Scholar, School of Computer Engineering and Technology, Dr. Vishwanath Karad MIT World Peace University, Pune- 411005, India*
[b] *Professor, School of Computer Engineering and Technology, Dr. Vishwanath Karad MIT World Peace University, Pune-411005, India*
[a*]*ruhina.karani@djsce.ac.in*

Abstract

Emotion recognition holds significant importance in human communication, benefiting various domains like human-computer interaction, affective computing, and social robotics. Recent interest lies in exploiting multimodal data, encompassing audio, visual, and other cues, to enhance emotion recognition systems. However, most available datasets predominantly focus on Western cultures, overlooking the diverse emotional expressions in regions like India. Moreover, existing datasets often neglect complex emotions like sympathy and awe. To address these limitations, we introduce "IndEMoVis," a novel multimodal dataset of Indian emotions. It comprises 122 recorded audio visual responses during conversations between pairs of individuals. The dataset includes 61 participants, consisting of 25 females and 36 males aged 18 to 21, primarily from Maharashtra and Gujarat states in India. It encompasses nine emotions: Neutral, Happiness, Sadness, Surprise, Disgust, Anger, Fear, Awe, and Sympathy. The annotation process involves a three-step procedure, ensuring accurate emotion labeling. Additionally, annotations are provided for intensity and confidence levels. IndEMoVis dataset aims to support the research community in affective computing by improving conversation abilities, analyzing emotional intelligence, and evaluating responses in debates. Its cultural relevance and inclusion of complex emotions offer valuable insights into emotion recognition for diverse contexts.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Indian Emotions; Dataset; Audio Visual Input; Multimodal Dataset; Emotion Recognition

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.200

1. Introduction

Emotions play a vital role in shaping behaviour, decision-making, and overall well-being, acting as powerful indicators of an individual's mental state, preferences, and intentions [1]. In today's interconnected society, the understanding of human emotions has grown in significance across various domains, including psychology, marketing, artificial intelligence, and human-computer interaction. Using artificial intelligence for emotion analysis offers valuable insights into customer sentiments, enabling personalized product and marketing strategies which is very useful in business sector. In healthcare sector using artificial intelligence as a tool, it shows promise in aiding the diagnosis and treatment of mental health conditions. Even though facial expressions are fundamental mode of communication, the research is shifting towards multimodal path for emotion recognition due to the advantages offered by individual modalities [2]. By combining facial expressions and auditory cues, these multimodal audio visual signals provide complementary and redundant information, leading to improved emotional clarity, cross-cultural communication, and emotion recognition accuracy. Moreover, they create more realistic and immersive experiences, foster emotional synchronization, and enhance accessibility for individuals with sensory impairments [2]. In the realm of e-commerce, AI-driven platforms analyze user behavior and spoken interactions during virtual shopping, decoding emotional responses through facial expressions and vocal cues. This informs highly personalized product recommendations, catering to diverse cultural backgrounds. In the healthcare sector, during teletherapy for patients dealing with anxiety and depression, a multimodal system assesses spoken words, facial expressions, and tone of voice. This comprehensive approach transcends language barriers, fostering empathetic cross-cultural therapeutic communication. These applications redefine the landscape of affective computing, adapting to our interconnected and culturally diverse world.

The evolution of emotion recognition techniques from controlled settings to real-world scenarios highlights the necessity for robust approaches. Deep learning methods exhibit promise in emotion recognition, but they demand substantial training data. Despite significant advancements in emotion recognition research, there remains a pressing need for comprehensive and culturally diverse multimodal datasets that encompass a wide range of emotions and real-world scenarios. Existing datasets often focus on single-modal data or have limited representation of emotion categories, particularly within specific cultural contexts.

In this paper, we propose a novel multimodal audio-visual dataset of Indian emotions, specifically designed to address the aforementioned limitations. Our dataset features audio-visual samples of individuals from the Indian population and includes the six basic emotion categories (Happiness, Sadness, Surprise, Disgust, Anger, Fear), along with complex emotion categories of Awe and Sympathy, and a Neutral emotion class. By including a broader spectrum of emotions, our dataset ensures greater accuracy and robustness in emotion recognition models tailored for Indian cultural nuances. Capturing emotions in controlled settings is insufficient to represent the dynamic nature of real-life emotional expressions. To overcome this, we collected dataset clips during conversational interactions between two participants, involving both speakers and listeners in spontaneous and guided communication scenarios. This approach introduces ecological validity and authenticity, making our dataset more reflective of genuine emotional responses in everyday interactions. To provide comprehensive and valuable insights into emotional expressions, our dataset is meticulously annotated not only for emotion categories but also for emotion intensity and confidence levels. Additionally, we annotate the dataset for the presence of braces, spectacles, and prominent hand gestures, which contribute to the contextual understanding of emotions and can significantly enhance emotion analysis models.

The availability of such a multimodal dataset of Indian emotions is expected to have profound implications in emotion recognition research and various practical applications. Researchers can use this dataset to develop and validate more accurate and culturally sensitive emotion recognition models, advancing the state-of-the-art in affective computing. Moreover, the dataset can facilitate the development of emotionally intelligent systems, enabling them to respond appropriately and empathetically to users' emotions.

The primary contributions of this research paper can be summarized as follows:

1. Proposal of a novel multimodal dataset featuring audio-visual samples of individuals from the Indian population, encompassing the six basic emotion categories (Happiness, Sadness, Surprise, Disgust, Anger, Fear) along with complex emotion categories (Awe and Sympathy) and a Neutral emotion class.

2. Collection of dataset clips during conversational interactions between two participants, incorporating both speakers and listeners, comprising spontaneous and guided communication scenarios.
3. Annotation of the dataset for emotion intensity and confidence levels in addition to the emotion categories. The dataset is also annotated for the presence of braces, spectacles, and prominent hand gestures, providing valuable context for emotion analysis.

The paper is structured as follows: Section 2 provides related work on existing datasets, Section 3 describes the proposed dataset, Section 4 provides results and discussions, Section 5 concludes the paper and discusses future work.

2. Related Work on Existing Emotion Datasets

The literature survey conducted reveals various datasets used for emotion recognition research, classified as single-modality and multimodal datasets.

Mollahosseini Ali et al. [3] introduced the Affectnet dataset, a single-modality dataset comprising facial images with annotated valence and arousal levels of emotions. The dataset includes seven distinct face emotions, namely Happy, Sad, Surprise, Fear, Disgust, Anger, and Contempt, manually noted in approximately half of the images (440K). Another dataset, ASCERTAIN, proposed by M. R. Subramanian et al. [4], is a multimodal dataset that incorporates data from various modalities such as electroencephalogram (EEG), electrocardiogram (ECG), galvanic skin response (GSR), and facial activity. This dataset includes information from 58 individuals with a mean age of 30 years, comprising 37 men and 21 women. P. Lucey et al. [5] suggested the Extended Cohn-Kanade (CK+) video dataset, which contains 593 video sequences from 123 distinct individuals, covering a wide age range (18 to 50) and diverse gender identities and ethnic backgrounds. The dataset assigns one of the seven expression classes, including Anger, Contempt, Disgust, Fear, Pleasure, Sorrow, and Surprise, to 327 of these video sequences. R. Kosti et al.[6] introduced the single-modality EMOTIC image dataset, which consists of 23,571 images annotated with dimensions for valence, arousal, and dominance. The dataset includes 26 emotion categories, such as Peace, Affection, Esteem, Anticipation, Engagement, Confidence, Happiness, Pleasure, Excitement, Surprise, Sympathy, Doubt/Confusion, Disconnection, Fatigue, Embarrassment, Yearning, Disapproval, Aversion, Annoyance, Anger, Sensitivity, Sadness, Disquietment, Fear, Pain, and Suffering. FER-2013 [7]is a proposed unimodal image dataset, consisting of 28,000 labeled images, divided into development, test, and training sets. The dataset includes seven emotions: Happy, Sad, Angry, Terrified, Surprised, and Disgusted.

The Google Facial Expression Comparison Dataset [8] is a single-modality dataset with triplets of facial expression images, annotated by human raters to identify which two faces in each triplet have the most comparable facial expressions. The dataset comprises 156K face photos and 500K to 156K triplets. Park, C.Y et al. [9] introduced K-EmoCon, a multimodal sensor dataset designed for continuous emotion recognition in naturalistic conversations. This dataset includes peripheral physiological signals, audiovisual recordings, and electroencephalogram. The multimodal audiovisual dataset eNTERFACE'05, introduced by O. Martin et al. [10], comprises 1166 video sequences with six emotions: Happiness, Sadness, Surprise, Fear, Disgust, and Anger. The participants in the dataset come from various countries, including Belgium, Cuba, Turkey, Slovakia, France, Brazil, Spain, the United States, Greece, Croatia, Italy, Canada, Austria, and Russia. Busso C. et al. [11] developed the multimodal and multilingual database known as IEMOCAP, which contains audiovisual material, including video, speech, face motion capture, and text transcriptions of 10 actors (5 men and 5 women). The dataset comprises dyadic sessions, featuring improvised scenes or scripted situations intentionally designed to elicit emotional reactions. Multiple annotators have labeled the IEMOCAP database with category labels such as "angry," "happy," "sad," and "neutral," as well as dimensional labels like "valence," "activation," and "dominance." JAFFE by Michael J. Lyons et al. [12,13] is a single-modality image dataset featuring 213 photographs of seven different facial expressions modeled by 10 Japanese women.

Soujanya Poria et al. [14] introduced MELD, a Multimodal Audio Visual Emotion Dataset comprising text, audio, and visual modalities from the TV series "Friends". The dataset includes more than 1400 lines and 13000 utterances, with assigned emotions such as Anger, Disgust, Sadness, Joy, Neutral, Surprise, and Fear. A multimodal library of speech and song known as RAVDESS, introduced by Livingstone SR and Russo FA [15], contains recordings of 24 professional actors performing lexically-matched phrases in a neutral North American accent. The dataset covers expressions like Peaceful, Pleased, Sad, Angry, Afraid, Surprised, and Disgusted. SEWA, a multimodal dataset by Kossaifi J. et al. [16], is a collection of annotated audio and 2D visual dynamic behavior data. It features recordings from six volunteer groups, each with 30 participants representing six distinct ethnic groups: British, German,

Hungarian, Greek, Serbian, and Chinese. The database includes more than 550 minutes of recorded computer-mediated face-to-face interactions between subject pairs and 1525 minutes of audio-visual data on people's responses to advertisements from 398 different subjects. The Indian Spontaneous Expression Database (ISED) [17] is a single-modality video dataset featuring Indian-natural face expressions. It includes 428 video clips from 50 healthy volunteers, comprising 29 men and 21 women, aged between 18 and 22, from various regions of India. The dataset includes four emotions: Happiness, Surprise, Sadness, and Disgust.

The IFExD dataset [18] offers a single-modality Image Dataset, encompassing 342 photos capturing eight distinct facial expressions. iSAFE [19] is a single-modality video collection, providing 395 clips of emotions, self-annotated by participants. IFED [20] introduces a single-modality video dataset, categorizing emotions into seven basic categories. Chen Jing et al. present HEU Emotion [21], a multimodal Video Emotion Dataset that includes ten emotions and two or three modalities. Additionally, the SAVEE dataset [22] offers a multimodal Audio Visual Dataset, featuring British English utterances from four male actors expressing six fundamental emotions and neutral emotions.

Based on an extensive literature review and the authors' knowledge, it is evident that all existing Indian emotion datasets are unimodal, containing facial image samples but lacking audio data. Notably, datasets like EMOTIC [6] and Google Facial Expression Comparison Dataset [8] include samples of complex emotions such as sympathy and awe, but they solely consist of unimodal facial image samples. Additionally, the existing Indian emotion datasets are obtained by eliciting responses from participants exposed to specific stimulant videos. In light of these observations, this research introduces a novel multimodal audio visual dataset encompassing samples of Indian emotions. The dataset is derived from spontaneous and guided communication interactions between two participants, covering the six basic emotion categories (Happiness, Sadness, Surprise, Disgust, Anger, Fear), as well as complex emotion categories (Awe and Sympathy), and a Neutral emotion class. Unlike existing datasets, the proposed dataset does not rely on stimulant videos to evoke emotions, making it more representative of real-world scenarios. By addressing the limitations of previous datasets and offering a multimodal perspective with audio visual samples, this research contributes to the advancement of emotion analysis and facilitates the development of more comprehensive and contextually relevant emotion recognition models.

3. Proposed Dataset

The dataset was created by capturing audio visual recordings of communication between two participants. While communicating the recordings were captured for speaker and listener both. This section describes the proposed dataset as well as the setup used for capturing these recordings, the participants engaged during the recordings and the approach used for annotation.

3.1 Data Collection Process

This section details the process followed during the creation of the dataset, encompassing the environmental setup, participant consent, and the annotation approach.

3.1.1 Creating an Environmental Setup

The data collection process for our multimodal audio-visual dataset of Indian emotions occurred within a controlled environment, involving pairs of participants engaging in conversations. To record the audio-visual responses of both participants, we utilized 13 MP mobile phone cameras placed at the centre of the table, facing each individual. The recorded audio-visual data was initially stored as raw recordings and later segmented into smaller clips with the assistance of participants to facilitate the self-annotation process.

3.1.2 Taking Participant Consent

A thorough consent protocol was followed before recording, which included outlining the educational, non-commercial purpose, acquiring participant demographics, and defining the data sharing and confidentiality restrictions. For record-keeping and anonymity, each participant was given a unique identification.

3.1.3 Recording of Data Samples

Mobile phone cameras were used to record the chats, which included both spontaneous and guided conversation. With the help of the participants, the raw data was later split into smaller video segments for easier management.

3.1.4 Annotating the Data Samples

We use a three-step method to identify emotion class annotations in the dataset. Participants begin by self-annotating based on their personal experiences. Then, using a proposed annotation algorithm that is thoroughly discussed in section 3.3, external annotators cross-annotate, and a final emotion category is established. A clinical psychologist checks the annotations for validity and clinical relevance in the third step.

3.2 Dataset Description

The "IndEMoVis" dataset is a rich resource that captures detailed conversational interactions between two individuals. In these conversations, one person takes on the role of the speaker, and the other person serves as the listener. Emotions can be expressed naturally throughout dialogue because to this unique setting. The dataset includes a variety of video recordings, including excerpts that highlight distinct emotions that were exhibited by the participants as well as raw videos of the full conversation and individual videos of the speakers and listeners. A thorough understanding of emotional expressions within the context of real conversation is provided by this enormous collection of video data. The dataset comprises a total of 122 raw videos, involving 61 participants, consisting of 25 females and 36 males, with ages ranging from 18 to 21 years. Each video captures a conversational interaction between two participants, with two cameras mounted on the table to record separate videos of the speaker and the listener. These recorded videos are subsequently segmented into clips, each showcasing different emotions expressed by the participants. The dataset encompasses nine distinct emotions, including Neutral, Happiness, Sadness, Surprise, Disgust, Anger, Fear, Awe, and Sympathy. The majority of participants are from Maharashtra and Gujarat states in India. The recordings captured a combination of spontaneous and guided conversations. In the spontaneous conversations, participants engaged in casual discussions with each other, while in the guided communication, participants were prompted to describe situations that elicited emotions from the different emotion categories mentioned earlier. This dual approach enriches the dataset, making it suitable for various research applications. The dataset consists of 45 recordings of guided conversations and 77 recordings of natural conversations, resulting in a total of 216 video clips portraying various emotions from the guided interactions and 379 video clips displaying emotions from the natural conversations. Additionally, the dataset offers annotations for emotion intensity and confidence levels, providing researchers with valuable insights into the strength and authenticity of emotional expressions. Figure 1 showcases selected samples for each emotion label, offering visual examples of the different emotional expressions captured in the dataset. The images are masked to protect the identity of participants. To facilitate further analysis and research, the video clips were further processed to extract image frames. From each clip, the frame with the highest intensity, capturing the peak expression of the corresponding emotion, was selected for subsequent analysis. These peak images serve as representative snapshots of the emotional expressions and can be used for facial emotion recognition tasks.

3.3 Data Annotation Process

The dataset's emotion category annotations undergo a comprehensive three-step process to ensure accuracy and reliability. Firstly, the participants self-annotate the video clips based on their emotional experiences. Subsequently, four independent external annotators cross-annotate the clips in the second step. The emotion category annotation is based on emotion similarity score generated by the GPT-3.5 Turbo Model between self-annotation emotion category and the emotion category annotated by each of the external annotators.

The total score for each video clip is computed using Equation I, where Ai represents the emotion annotated by an external annotator, and P represents the emotion self-annotated by the participant. The AiP similarity score falls within the range of 0 to 1, serving as a measure of similarity between the emotion category annotated by an external

annotator and the emotion category self-annotated by the participant. This score is determined through the utilization of the ChatGPT turbo model, taking into account various factors, including semantics, potential confusion, expressiveness of emotions, and the individual's role in the interaction. It is noteworthy that a score of 1 signifies a perfect match between the two emotion categories, while a score of 0 indicates complete dissimilarity. It's essential to emphasize that the ChatGPT model relies on a qualitative approach for this computation, drawing from its understanding of general linguistic patterns and the corpus it was trained on. As an illustration, the model associates "sympathy" with "sadness" due to the common expression of sympathy when an individual is experiencing sadness, whereas happiness and sadness are generally regarded as contrasting emotions, aligning with conventional linguistic patterns.

$$Total\ Score = \sum_{i=1}^{4} A_i P \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (I)$$

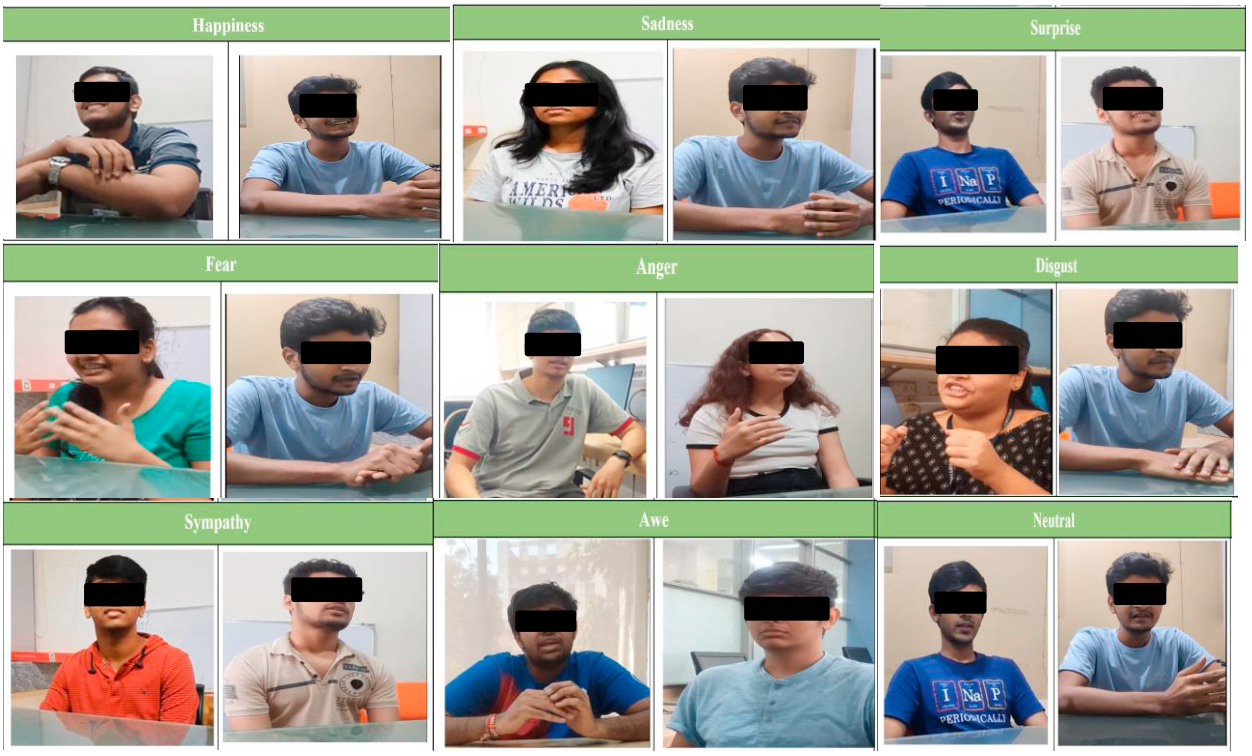


Fig. 1. Samples of each Emotion Label

Algorithm I is employed to determine the final annotation of emotion classes in the dataset. A threshold value of 3 is set, representing a majority agreement point beyond the midpoint value. If the total score falls below this threshold, indicating significant disparity in the emotion categories annotated by external annotators, the self-annotations by participants are accepted as the final annotations. On the other hand, if the total score equals or exceeds the threshold, suggesting high similarity among external annotators and the participant, the final annotation is randomly selected with a probability of 0.5 from either the majority-voted emotion among external annotators or the emotion self-annotated by the participant.

In the third and final step, the annotations generated by the algorithm are re-verified by an experienced clinical psychologist to ensure their validity and clinical relevance. Additionally, the dataset is annotated for the presence of braces, spectacles, and prominent hand gestures, providing valuable context for emotion analysis. Furthermore, intensity and confidence levels are annotated on a scale of 0 to 5 and 0 to 7, respectively, by the clinical psychologist to further enrich the dataset with nuanced emotional expressions.

Algorithm I: Algorithm for Final Annotation of Dataset for Emotion Categories

```
1) Calculate total score using Annotator-Participant similarity score as per equation II.
2) If Total Score < 3 (Max Threshold decided)
      Then Final Annotation = Self Annotation
3) If Total Score >=3 (Max Threshold decided)
      Perform coin toss event using random number r (between 0 and 1)
      If (r > 0.5):
         Then Final Annotation = Self Annotation
      Else:
          If (Max voting for single emotion category by External Annotators)
              Then Final Annotation = Max Voted Emotion Category
          Else (tiebreaker)
              Final Annotation = emotion category of highest similarity score (between external annotators and self-annotator)
```

4. Results and Discussions

In this section, we present the results of our comparative analysis between the proposed multimodal audio-visual dataset and existing Indian emotion datasets, the exploratory data analysis and results of annotation algorithm. This discussion underscores the significance of our contribution, setting the stage for a comprehensive exploration of the implications of our research in the field of emotion analysis.

4.1 Comparative Analysis of the Proposed Multimodal Audio-Visual Dataset

Our study involved a thorough examination of the proposed multimodal audio-visual dataset in contrast to existing Indian emotion datasets. Drawing upon an extensive literature review and our own domain knowledge, it becomes evident that the existing Indian emotion datasets predominantly follow a unimodal approach. They are primarily comprised of facial image samples, lacking the incorporation of audio data, which is critical for a comprehensive understanding of emotions. Significantly, renowned datasets such as EMOTIC [6] and Google Facial Expression Comparison Dataset [8] have made strides in including samples of complex emotions like sympathy and awe. However, these datasets, too, remain rooted in unimodal facial image samples. Additionally, the prevailing approach in existing Indian emotion datasets involves eliciting responses from participants exposed to specific stimulant videos. This methodology, while useful in controlled settings, may not faithfully represent the complexities of genuine, real-world emotional expressions.

Our research, on the other hand, breaks away from the unimodal tradition and introduces a pioneering multimodal audio-visual dataset, which comprehensively encapsulates the spectrum of Indian emotions. It is derived from both spontaneous and guided communication interactions between two participants and includes the six basic emotion categories (Happiness, Sadness, Surprise, Disgust, Anger, Fear), as well as complex emotion categories (Awe and Sympathy), and a Neutral emotion class. Importantly, our dataset deliberately avoids the use of stimulant videos, opting for authentic, real-world scenarios that evoke genuine emotional responses. This critical shift towards a multimodal perspective, where both facial expressions and auditory cues are considered, defines the essence of our contribution. It illuminates the intrinsic limitations of existing datasets and sets a new standard for emotion analysis within an Indian context. The integration of audio data provides a more comprehensive and contextually relevant understanding of emotions, greatly enhancing the dataset's practical utility across diverse fields, including psychology, marketing, artificial intelligence, and human-computer interaction.

4.2 Exploratory Data Analysis

Table 1 provides an overview of the number of video clip samples collected for each emotion category, while Figure 2 visually illustrates the distribution of these samples across the various emotions, highlighting the relative proportions of each emotion category. The dataset's emotions distribution is noticeably varied, with "Neutral," which

is represented by 156 samples, being the most common emotional state. With 114 samples, "happiness" comes in second, strongly suggesting the presence of happy feelings. 'Sadness,' 'Surprise,' 'Fear,' and 'Sympathy' are also well represented, with 67, 72, 50, and 49 samples, respectively, providing a rich depiction of diverse emotional states. Additionally, the dataset contains 34 and 33 examples, respectively, for the emotions of "disgust" and "anger." With 20 samples, it is clear that complex emotions like "Awe" have been included. The dataset will accurately reflect a range of emotional expressions because to its balanced distribution. The findings also show that the two emotions that are most frequently expressed in these conversations are neutral and happy. On the other hand, emotions like Awe and Disgust are relatively less frequently exhibited.

The emotional expressions observed in both the guided and spontaneous conversations contribute to a comprehensive set of emotion samples in the dataset. Table 2 provides a detailed breakdown of the number of video clips for each emotion category in both the guided and spontaneous recordings. Additionally, Figure 3 presents the percentage-wise distribution of emotions captured in the raw recordings and the corresponding video clips.

In total, there are 122 raw video recordings, with 77 capturing spontaneous emotions and 45 featuring guided emotions. The dataset's authenticity is built on these raw recordings, which show actual emotional responses in both guided and naturally occurring encounters. A total of 595 video clips, including 379 instances of spontaneous emotion and 216 instances of guided emotion, are added to the dataset to further enhance it. The numerous video clips available demonstrate the segmentation process, which divides the original recordings into more manageable chunks for analysis. The dataset is a useful tool for study and applications in emotion analysis, human-computer interaction, and affective computing since its rigorous categorization enables a detailed exploration of emotional expressions. The dataset's depth and authenticity are enhanced by the inclusion of both spontaneous and guided emotions, which guarantees that a wide range of emotional experiences is captured.

Table 1: Summary of number of samples created for each emotion

Sr. No.	Emotions	No. of Samples Collected
1.	Neutral	156
2.	Happiness	114
3.	Sadness	67
4.	Surprise	72
5.	Disgust	34
6.	Anger	33
7.	Fear	50
8.	Awe	20
9.	Sympathy	49
Total		595

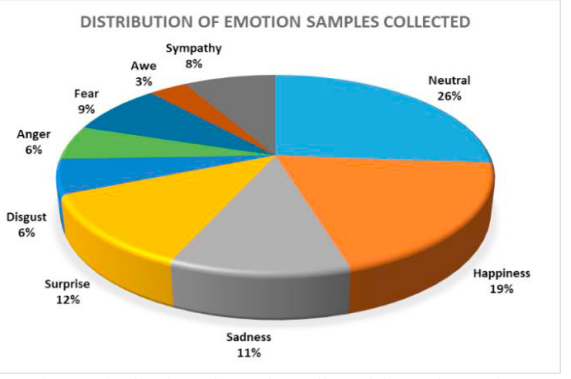


Fig. 2. Distribution of samples collected for each emotion

Table 2. Bifurcation of Spontaneous and Guided Emotions

	Spontaneous Emotions	Guided Emotions	Total
Raw video Recordings	77	45	122
Video Clips	379	216	595

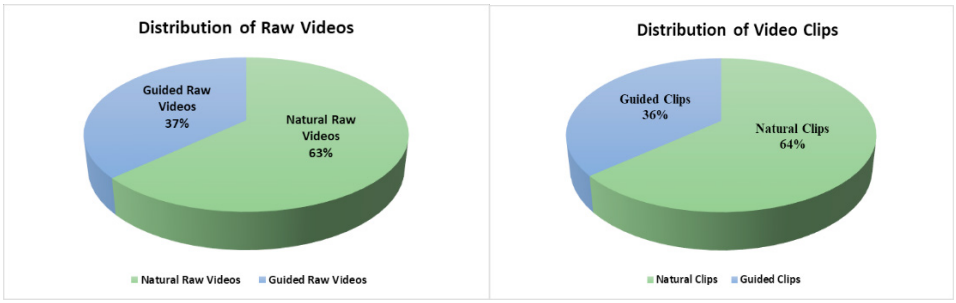


Fig. 3. Percentage wise Distribution of Guided and Spontaneous Emotions

4.3 Results of annotation algorithm

This section describes the annotation result achieved by applying Algorithm I. The Generative Pre-Trained Transformers (GPT-3.5) Turbo model is utilized to generate the emotion similarity score for each Annotator-Participant pair, taking into account both the speaker and the listener roles played by the participant. It is important to note that the external annotator always assumes the role of a listener, as they annotate the recordings based on their perception. Table 3 illustrates the samples of similarity scores generated when the participant acts as a speaker, and the annotator serves as a listener. Conversely, Table 4 presents the samples of similarity scores when both the participant and the annotator assume the role of listeners. Figure 4 describes the samples of final annotation of proposed dataset after undergoing a three step annotation process.

Table 3. Similarity Scores generated when Participant is a speaker and Annotator is a listener

Emotion Category Annotation		Similarity Score
Participant	Annotator	
Happy	Happy	1
Happy	Sad	0.6
Happy	Fear	0.4
Happy	Anger	0.3
Happy	Neutral	0.8
Happy	Surprise	0.7
Happy	Sympathy	0.8
Happy	Awe	0.6

Table 4. Similarity Scores when Both Participant and Annotators are listeners

Emotion Category Annotation		Similarity Score
Participant	Annotator	
Happy	Sad	0.65
Happy	Fear	0.4
Happy	Anger	0.3
Happy	Neutral	0.75
Happy	Surprise	0.6
Happy	Sympathy	0.8
Happy	Awe	0.5
Happy	Disgust	0.3

ID	Role	Type of Clips	Clip1	Intensity of Expression	Confidence	Clip2	Intensity of Expression	Confidence	Clip3	Intensity of Expression	Confidence	Clip4	Intensity of Expression	Confidence
1	Listener	Natural	Neutral			Surprise	3	5	Sympathy			NA		
2	Listener	Natural	Neutral			Happy	4	4	Surprise	4	4	Happy	4	4
3	Listener	Natural	Neutral			Surprise	2	4	NA			NA		
4	Speaker	Natural	Neutral			Fear	2	5	Happy			NA		
5	Listener	Natural	Neutral			Happy	4	6	NA			NA		
6	Speaker	Natural	Neutral			Fear	4	4	NA			NA		
7	Speaker	Natural	Neutral			Happy	4	4	NA			NA		
8	Speaker	Natural	Neutral			Happy	5	7	NA			NA		
9	Speaker	Natural	Happy	3	5	NA			NA			NA		
10	Listener	Natural	Neutral			Neutral			Surprise	2	4	NA		

Fig. 4. Samples of final annotations

4.4 Dataset Availability

The dataset is available on the link https://doi.org/10.5281/zenodo.8255783. It can be requested by signing an End User License Agreement.

5. Conclusion and Future Work

In this paper, we proposed a novel multimodal audio-visual dataset of Indian emotions, IndEmoVis having six basic emotion categories Happiness, Sadness, Surprise, Disgust, Anger and Fear, along with complex emotion categories of Awe and Sympathy, and a Neutral emotion class. We captured clips in real conversational interactions involving speakers and listeners, fostering authenticity. This approach introduced ecological validity and authenticity, making our dataset more reflective of genuine emotional responses in everyday interactions. To provide comprehensive and valuable insights into emotional expressions, the dataset not only annotated emotion categories but also included emotion intensity, confidence levels, and detected elements like braces, spectacles, and prominent hand gestures. The availability of such a multimodal dataset of Indian emotions is expected to have profound implications in emotion recognition research and various practical applications. Researchers can use this dataset to develop and validate more accurate and culturally sensitive emotion recognition models, advancing the state-of-the-art in affective computing. The future work is aimed at enriching this dataset with more samples taken from real time environment capturing the conversation between the group of people and also applying various techniques to set the baseline for the proposed dataset.

References

[1]. Tyng Chai M., Amin Hafeez U., Saad Mohamad N. M. and Malik Aamir S., " The Influences of Emotion on Learning and Memory ", Frontiers in Psychology, Volume 8, 2017, https://www.frontiersin.org/articles/10.3389/fpsyg.2017.01454

[2]. Ruhina Karani and Sharmishta Desai, “Review on Multimodal Fusion Techniques for Human Emotion Recognition” International Journal of Advanced Computer Science and Applications (IJACSA), 13(10), 2022. http://dx.doi.org/10.14569/IJACSA.2022.0131035

[3]. A. Mollahosseini, B. Hasani and M. H. Mahoor, "AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild," in IEEE Transactions on Affective Computing, vol. 10, no. 1, pp. 18-31, 1 Jan.-March 2019, doi: 10.1109/TAFFC.2017.2740923.

[4]. R. Subramanian, J. Wache, M. K. Abadi, R. L. Vieriu, S. Winkler and N. Sebe, "ASCERTAIN: Emotion and Personality Recognition Using Commercial Sensors," in IEEE Transactions on Affective Computing, vol. 9, no. 2, pp. 147-160, 1 April-June 2018, doi: 10.1109/TAFFC.2016.2625250.

[5]. P. Lucey, J. F. Cohn, T. Kanade, J. Saragih, Z. Ambadar and I. Matthews, "The Extended Cohn-Kanade Dataset (CK+): A complete dataset for action unit and emotion-specified expression," 2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition - Workshops, San Francisco, CA, USA, 2010, pp. 94-101, doi: 10.1109/CVPRW.2010.5543262.

[6]. R. Kosti, J. M. Alvarez, A. Recasens and A. Lapedriza, "EMOTIC: Emotions in Context Dataset," 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Honolulu, HI, USA, 2017, pp. 2309-2317, doi: 10.1109/CVPRW.2017.285.

[7]. I. J. Goodfellow, D. Erhan, P. L. Carrier, A. Courville, M. Mirza, B. Hamner, W. Cukierski, Y. Tang, D. Thaler, D.-H. Lee, Y. Zhou, C. Ramaiah, F. Feng, R. Li, X. Wang, D. Athanasakis, J. Shawe-Taylor, M. Milakov, J. Park, R. Ionescu, M. Popescu, C. Grozea, J. Bergstra, J. Xie, L. Romaszko, B. Xu, Z. Chuang, and Y. Bengio, "Challenges in representation learning: A report on three machine learning contests," in Neural Networks, vol. 64, pp. 59-63, 2015.

[8]. R. Vemulapalli and A. Agarwala, "A Compact Embedding for Facial Expression Similarity," 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA, 2019, pp. 5676-5685, doi: 10.1109/CVPR.2019.00583.

[9]. C. Y. Park, N. Cha, S. Kang, A. Kim, A. H. Khandoker, L. Hadjileontiadis, A. Oh, Y. Jeong and U. Lee, "K-EmoCon, a multimodal sensor dataset for continuous emotion recognition in naturalistic conversations." Sci Data 7, vol. 293, 2020.

[10]. I. Pitas, I. Kotsia, O. Martin and B. Macq, "The eNTERFACE?05 Audio-Visual Emotion Database," in 2006 22nd International Conference on Data Engineering Workshops, Atlanta, GA, USA, 2006 pp. 8. doi: 10.1109/ICDEW.2006.145

[11]. C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee and S. S. Narayanan, " IEMOCAP: interactive emotional dyadic motion capture database," Lang Resources & Evaluation, vol. 42, no. 4, 2008, pp. 335–359, https://doi.org/10.1007/s10579-008-9076-6.

[12]. M. Lyons, S. Akamatsu, M. Kamachi and J. Gyoba, "Coding facial expressions with Gabor wavelets," Proceedings Third IEEE International Conference on Automatic Face and Gesture Recognition, Nara, Japan, 1998, pp. 200-205, doi: 10.1109/AFGR.1998.670949.

[13]. M. Lyons, "Excavating AI" Re-Excavated: Debunking a Fallacious Account of the Jaffe Dataset, July 28, 2021. [Online]. Available: https://ssrn.com/abstract=3900990 or http://dx.doi.org/10.2139/ssrn.3900990.

[14]. S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations," in Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Florence, Italy, 2019, pp. 527-536, https://aclanthology.org/P19-1050/.

[15]. S. R. Livingstone and F. A. Russo, "The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," PLoS ONE, vol. 13, no. 5, pp. 1-9, 2018, https://doi.org/10.1371/journal.pone.0196391

[16]. J. Kossaifi, R. Walecki, Y. Panagakis, J. Shen, M. Schmitt, F. Ringeval, J. Han, V. Pandit, A. Toisoul, B. Schuller, K. Star, E. Hajiyev and M. Pantic, "Sewa db: A rich database for audio-visual emotion and sentiment research in the wild," IEEE transactions on pattern analysis and machine intelligence, vol. 43, no. 3, pp. 1022-1040, 1 March 2021, doi: 10.1109/TPAMI.2019.2944808.

[17]. S. L. Happy, P. Patnaik, A. Routray, and R. Guha, "The Indian Spontaneous Expression Database for Emotion Recognition," in IEEE Transactions on Affective Computing, vol. 8, no. 1, pp. 131-142, Jan.-Mar. 2017, doi: 10.1109/TAFFC.2015.2498174

[18]. IFExD, Indian Facial Expression Dataset. [Online]. Available: https://ifexd.github.io/index, Accessed on October 11, 2023.

[19]. S. Singh and S. Benedict, "Indian Semi-Acted Facial Expression (iSAFE) Dataset for Human Emotions Recognition," in Advances in Signal Processing and Intelligent Recognition Systems (SIRS 2019), S. Thampi, et al., Eds., Communications in Computer and Information Science, vol. 1209, Springer, Singapore, 2020, pp. 176-186, doi: 10.1007/978-981-15-4828-4_13.

[20]. Arunashri, V. K. V. Venkateshu, and C. Lalitha, "A database for facial expressions among Indians," MedPulse International Journal of Anatomy, vol. 17, no. 2, pp. 12-16, February 2021. [Online]. Available: http://www.medpulse.in/Anatomy.

[21]. J. Chen, C. Wang, K. Wang, et al., "HEU Emotion: a large-scale database for multimodal emotion recognition in the wild," Neural Computing and Applications, vol. 33, no. 17, pp. 8669-8685, 2021, https://doi.org/10.1007/s00521-020-05616-w

[22]. P. J. B. Jackson and S. u. haq, Surrey Audio-Visual Expressed Emotion (SAVEE) database, 2011, [Online]. Available: http://kahlan.eps.surrey.ac.uk/savee/